Lecture 24
Tests for p and μ

Review

1 **A drug maker says 80% of patients improve. A doctor suspects fewer do. State H_0 and H_A.**

H_0: $p = 0.80$ H_A: $p < 0.80$ ($p_0 = 0.80$, lower-tailed)

2 **What is the test statistic z_{obs}?**

How many SEs the estimate is from θ_0: $z_{obs} = (\text{estimate} - \theta_0) / \text{SE}$

3 **What is the P-value?**

The probability of a result as or more extreme than z_{obs}, if H_0 is true

Making a decision

significance level α

the cutoff for "rare enough", chosen before collecting data; usually $\alpha = 0.05$

P-value $< \alpha$

reject H_0: the result is statistically significant, evidence for H_A

P-value $\geq \alpha$

do not reject H_0: the data are not convincing evidence for H_A

"do not reject" is not "accept"

not rejecting H_0 does not prove it true; the evidence just isn't strong enough

Practice: P-values and decisions

For each test, write the P-value as a probability given H_0 is true. Then use its value to make a decision.

P-value = P(? | H_0 true)

(a)	**H_0: p = 0.40**	**H_A: p > 0.40**	**z_{obs} = 1.52**	**α = 0.05**	**P-value = 0.0643**
(b)	**H_0: μ = 12**	**H_A: μ < 12**	**t_{obs} = −2.30, df = 15**	**α = 0.01**	**P-value = 0.0181**
(c)	**H_0: μ = 50**	**H_A: μ ≠ 50**	**z_{obs} = −2.10**	**α = 0.05**	**P-value = 0.0358**

(a) P(Z ≥ 1.52 | H_0 true) = 0.0643 ≥ 0.05: do not reject H_0
(b) P(T ≤ −2.30 | H_0 true) = 0.0181 ≥ 0.01: do not reject H_0
(c) P(Z ≤ −2.10 or Z ≥ 2.10 | H_0 true) = 2(0.0179) = 0.0358 < 0.05: reject H_0

Testing a population proportion

when to use it

a categorical variable, and a claim H_0: $p = p_0$ about the proportion of successes

assumptions

a random sample, and $np_0 \geq 15$ and $n(1 - p_0) \geq 15$

test statistic

$z_{obs} = (\hat{p} - p_0) / \sqrt{(p_0(1 - p_0) / n)}$: the SE uses p_0, because we assume H_0 is true

P-value

upper: $P(Z \geq z_{obs} \mid H_0 \text{ true})$; lower: $P(Z \leq z_{obs} \mid H_0 \text{ true})$; two-sided: double the tail

Example: cell phones and brain cancer

In a study of 420,019 cell phone users, 172 developed brain cancer. For people who do not use cell phones, the rate is 0.034%. Do cell phone users get brain cancer at a higher rate? (OpenStax, Example 9.21)

$x = 172$; $n = 420{,}019$; $p_0 = 0.00034$; $\alpha = 0.005$

(a) What are H_0 and H_A? Which tail? Are the assumptions met?
(b) What are $\hat{p}$, the standard error and z_{obs}?
(c) What is the P-value? What is the decision at $\alpha = 0.005$?
(d) Why would we choose such a small α here?

(a) H_0: $p = 0.00034$ H_A: $p > 0.00034$, upper-tailed; $np_0 = 142.8 \geq 15$, $n(1 - p_0) \geq 15$
(b) $\hat{p} = 172 / 420{,}019 \approx 0.0004095$; $SE = \sqrt{p_0(1 - p_0) / n} \approx 0.00002845$
$z_{obs} = (0.0004095 - 0.00034) / 0.00002845 \approx 2.44$
(c) $P(Z \geq 2.44 \mid H_0 \text{ true}) = 1 - 0.9927 = 0.0073 \geq 0.005$: do not reject H_0
(d) Rejecting a true H_0 would wrongly blame phones for cancer, so we ask for strong evidence

Testing a population mean

when to use it

a claim H_0: $\mu = \mu_0$; a random sample from a normal population, or $n \geq 30$

σ known: z-test

$z_{obs} = (\bar{x} - \mu_0) / (\sigma/\sqrt{n})$, P-value from the z-table

σ unknown: t-test

$t_{obs} = (\bar{x} - \mu_0) / (s/\sqrt{n})$, df = n − 1, P-value from the t-table

P-value from the t-table

find t_{obs} between two values in row df; the P-value is between their tail areas

Example: the first statistics test

Students believe the mean score on the first statistics test is 65. An instructor thinks it is higher and samples 10 students' scores, which are normal. (OpenStax, Example 9.16)

$n = 10$; $\bar{x} = 67$; $s = 3.197$; $\mu_0 = 65$; $\alpha = 0.05$

(a) What are H_0 and H_A? Which tail? Are the assumptions met?
(b) z or t? What are the standard error and the test statistic?
(c) What is the P-value (use the t-table)? What is the decision?

(a) H_0: $\mu = 65$ H_A: $\mu > 65$, upper-tailed; $n < 30$, but the scores are normal
(b) σ unknown, so t; $SE = 3.197 / \sqrt{10} \approx 1.011$; $t_{obs} = (67 - 65) / 1.011 \approx 1.98$, $df = 9$
(c) row 9: $1.833 < 1.98 < 2.262$, so $0.025 <$ P-value < 0.05: reject H_0

Practice: younger than the groom?

Joon believes 50% of first-time brides in the U.S. are younger than their grooms. She tests whether the percentage is different from 50%. (OpenStax, Example 9.17)

n = 100 brides; 53 younger than their grooms; $p_0 = 0.50$; $\alpha = 0.01$

(a) Are the assumptions met?
(b) What are H_0 and H_A? Which tail?
(c) What are $\hat{p}$, the standard error and z_{obs}?
(d) What is the P-value? What is the decision?

(a) a random sample; $np_0 = 100(0.50) = 50 \geq 15$ and $n(1 - p_0) = 50 \geq 15$
(b) H_0: $p = 0.50$ H_A: $p \neq 0.50$; two-sided
(c) $\hat{p} = 53 / 100 = 0.53$; $SE = \sqrt{(0.50 \times 0.50 / 100)} = 0.05$; $z_{obs} = 0.03 / 0.05 = 0.60$
(d) $2 \times P(Z \geq 0.60 \mid H_0 \text{ true}) = 2(1 - 0.7257) = 0.5486 \geq 0.01$: do not reject H_0

Practice: faster with goggles?

Jeffrey swims the 25-yard freestyle in a mean of 16.43 s, with σ = 0.8 s. In new goggles, he swims it 15 times. Do the goggles make him faster? Times are normal. (OpenStax, Example 9.14)

$n = 15$; $\bar{x} = 16$ s; $\sigma = 0.8$ s; $\mu_0 = 16.43$ s; $\alpha = 0.05$

(a) What are H_0 and H_A? Which tail?
(b) z or t? What are the standard error and the test statistic?
(c) What is the P-value? What is the decision?

(a) H_0: $\mu = 16.43$ H_A: $\mu < 16.43$, lower-tailed: faster means less time
(b) σ known, so z; $SE = 0.8 / \sqrt{15} \approx 0.2066$; $z_{obs} = (16 - 16.43) / 0.2066 \approx -2.08$
(c) $P(Z \le -2.08 \mid H_0 \text{ true}) = 0.0188 < 0.05$: reject H_0; the goggles help